

Project Report

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Project - 1

Name of the Project: Foot Printing

Tools Used: *(Mention the tools used in this project)*

- Netcraft and Nmmapper

Working Procedure: *(Answer in 200 words with bullet points)*

- Netcraft tools are widely used for monitoring and analysing websites, particularly for identifying vulnerabilities and security risks.
- Visit the Netcraft website in a browser and navigate to the "Site Report" service
- The following is the website used for report : <https://codepen.io>
- Enter the URL and click "Get Report" to start the analysis.
- Netcraft gets the details of site background , hosting details, SSL certificates and potential security issues.
- Below is some of the results of running the site report on
- " <https://codepen.io> "

Background

Site title	Just a moment...	Date first seen	June 2012
Site rank	964	Primary language	English
Description	Not Present		

Network

Site	https://codepen.io	Domain	codepen.io
Netblock Owner	Cloudflare, Inc.	Nameserver	albert.ns.cloudflare.com
Hosting company	Cloudflare	Domain registrar	Unknown
Hosting country	US	Nameserver organisation	whois.cloudflare.com
IPv4 address	104.18.111.164	Organisation	Unknown
IPv4 autonomous systems	AS13335	DNS admin	dns@cloudflare.com
IPv6 address	2606:4700:0:0:0:0:6810:20e4	Top Level Domain	British Indian Ocean Territory (.io)
IPv6 autonomous systems	AS13335	DNS Security Extensions	Enabled
Reverse DNS	Unknown		

- Nmmapper is a web-based tool similar to Netcraft which is used to identify and map subdomains of a given domain which can be crucial for security audits, penetration testing and gaining insights into a company's online infrastructure
- Visit Nmmapper website and enter the domain for which we want to run the same.
- Click on "search" to start the process of subdomain discovery.

- The page results in the list of discovered subdomains associated with the primary domain.
- To read further details , click on the individual subdomains to view detailed DNS information.
- The following are the results for running Nmmapper for the website "https://codepen.io":

Found 65 subdomains, but returning 15. Become a patron or buy us a coffee and Signup to INCREASE returned subdomains

Host	Subdomain	IP	ASN
codepen.io	webui_84033452.codepen.io		
codepen.io	5pz7vmukms.this_is_research.next_query.is_from_us.if_problem.please_see.http.researchscan.cs.berkeley.edu.codepen.io		
codepen.io	assets.codepen.io		
codepen.io	mail.codepen.io		
codepen.io	maller.codepen.io		
codepen.io	jzdpwyjmrg.this_is_research.next_query.is_from_us.if_problem.please_see.http.researchscan.cs.berkeley.edu.codepen.io		
codepen.io	2rnhxyrkr.this_is_research.next_query.is_from_us.if_problem.please_see.http.researchscan.cs.berkeley.edu.codepen.io		
codepen.io	futures.codepen.io		
codepen.io	webui_39263941.codepen.io		
codepen.io	9yhfmza44d.this_is_research.next_query.is_from_us.if_problem.please_see.http.researchscan.cs.berkeley.edu.codepen.io		
codepen.io	production.etl.codepen.io		
codepen.io	podcast.codepen.io		
codepen.io	llmbr.codepen.io		

Learning Outcomes: (Answer in 100 words with bullet points)

- Using Netcraft tool can help gain insights into the hosting details, server configurations and technologies used by websites
- We can also know the potential vulnerabilities, outdated software and phishing threats through detailed reports.
- Nmmapper can be used to learn to identify and map subdomains associated with a primary domain , a key skill in reconnaissance and security audits
- The results can be used for understanding how to analyze subdomains for potential vulnerabilities and misconfigurations
- For comprehensive analysis , we can also export the discovered data from the results and integrate it with other security tools

Project - 2

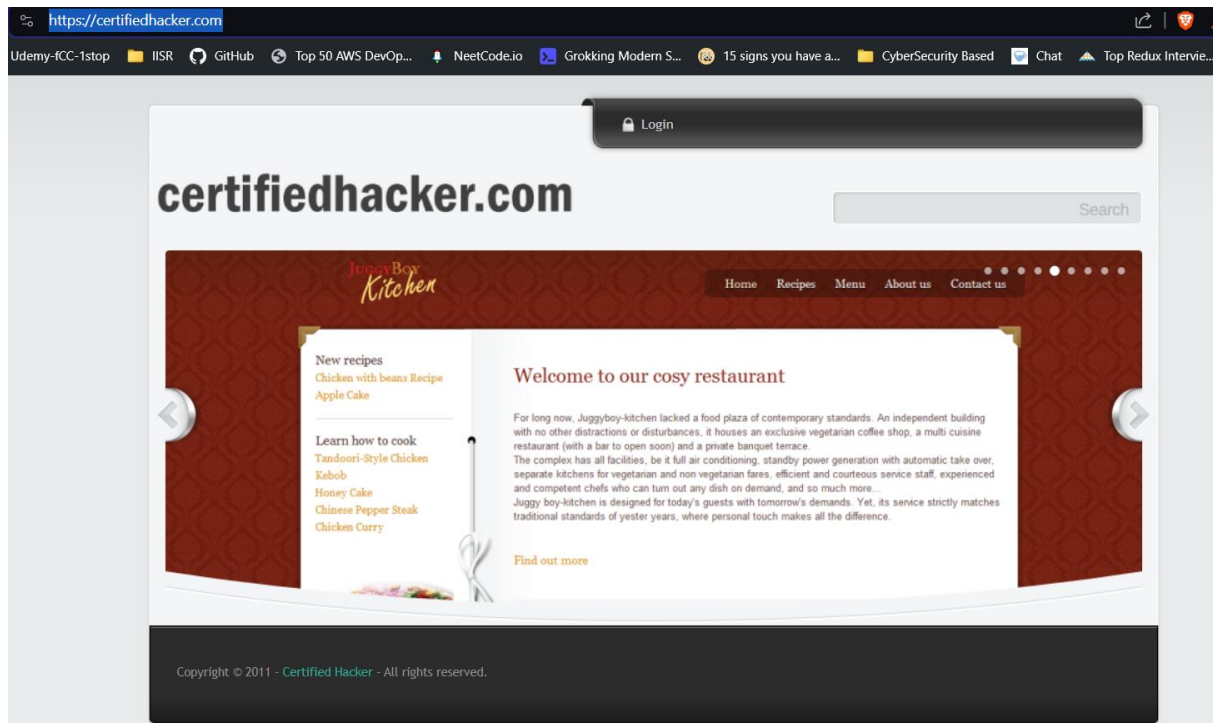
Name of the Project: Mirroring of Websites

Tools Used: *(Mention the tools used in this project)*

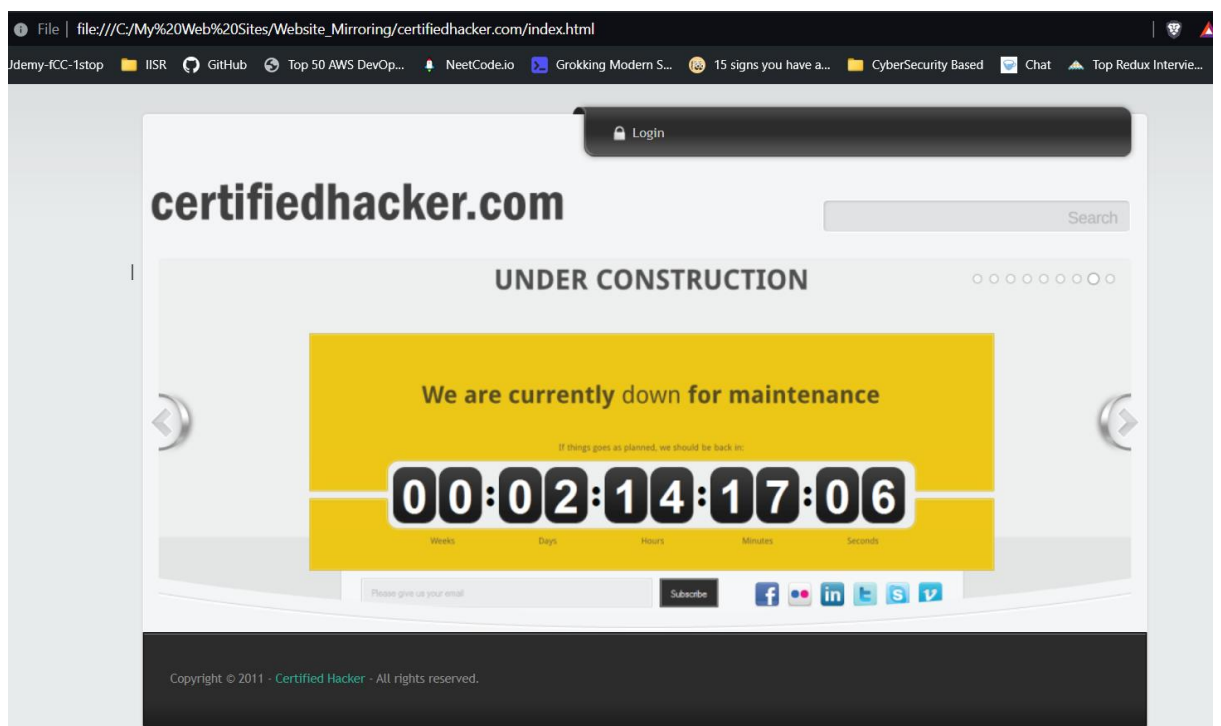
- HTTRACK Website Copier

Working Procedure: *(Answer in 200 words with bullet points)*

- HTTrack Website Copier is an offline browser utility that allows users to download a website from the Internet to a local directory, making all directories getting HTML, images and other files from the server to the user's computer.
- Firstly, we need to install the application from the official HTTrack website that is suitable to the system OS
- Launch the Application
- Click on "Next" in the opening page to create a New Project. Enter the name of the new Project
- Select the base path where you want to save the project and Enter the website URL that you want to download
- For this website, I have used a test website <https://certifiedhacker.com/>
- Click on Next and then Finish to start the download process.
- HTTrack will crawl through the website and gather all the required files to the local directory where we specified.
- The process duration depends on the size of the website
- Once the download is completed ,visit the files in your local storage where they are saved.
- Open the "index.html" file in the web browser to view the offline version of the website you have downloaded .All internal links are also updated to the local files.
- Compare the Original to the downloaded website
- Original Website:



- Mirrored Website:



Learning Outcomes: (Answer in 100 words with bullet points)

- Using HTTrack for mirroring websites can help us understand the process of downloading and viewing websites offline which is useful for analysis(website construction) and archiving(in case new updates of the same website are released , we can compare the two for research purpose)
- We can gain insights into the structure and content organization of websites

- We can also configure the download settings to include or exclude specific content , manage download limits and handle dynamic content
- Downloading websites can help us change the structure or any other changes and save them permanently

Project - 3

Name of the Project: Creating a New user in Kali

Tools Used: *(Mention the tools used in this project)*

- Virtual Box ,Kali Linux

Working Procedure: *(Answer in 200 words with bullet points)*

- For this project we need to first install virtual box and Kali Linux in virtual box
- Visit the official website of Oracle VM Virtualbox and download the latest version of Virtualbox on your OS
- Follow the Installation process of the same and install it
- Next download the Kali Linux latest version from the official website and follow the installation process to install it in the virtual box
- Run the Kali Linux in Virtualbox with the username and password (default:kali for both)
- In Kali linux add another account of an user where we will be installing Tor
- For adding a new user , open the terminal in kali and run the following command:
 - Sudo usermod -aG sudo user2
 - This will ask for kali password to authenticate the action, Type the password to start the password
 - It will ask for the password user2 ,give the password for the user and continue
 - (Remember the username and password)
- To be able to use sudo privileges by user2 run the following commands:
 - sudo visudo
 - This will open the sudoers file
- Go to the bottom of the file and type:
 - user2 ALL=(ALL:ALL) ALL
 - This gives all the sudo privileges to user2
 - Save and exit
- To verify the changes , Switch the user from default -kali to user2 and give the password assigned for the same
- Run the command : "sudo apt update" to update the changes in user2

Learning Outcomes: *(Answer in 100 words with bullet points)*

- Learning the process of creating user accounts can help understand the user management in a linux environment which is crucial for system administration.
- Gain knowledge about user permissions and access controls in kali linux
- Enhance command line skills including commands like `adduser` and `usermod` for user management
- Understand the importance of creating separate user accounts for different tasks to enhance security.

Project - 4

Name of the Project: Configuring Proxychains

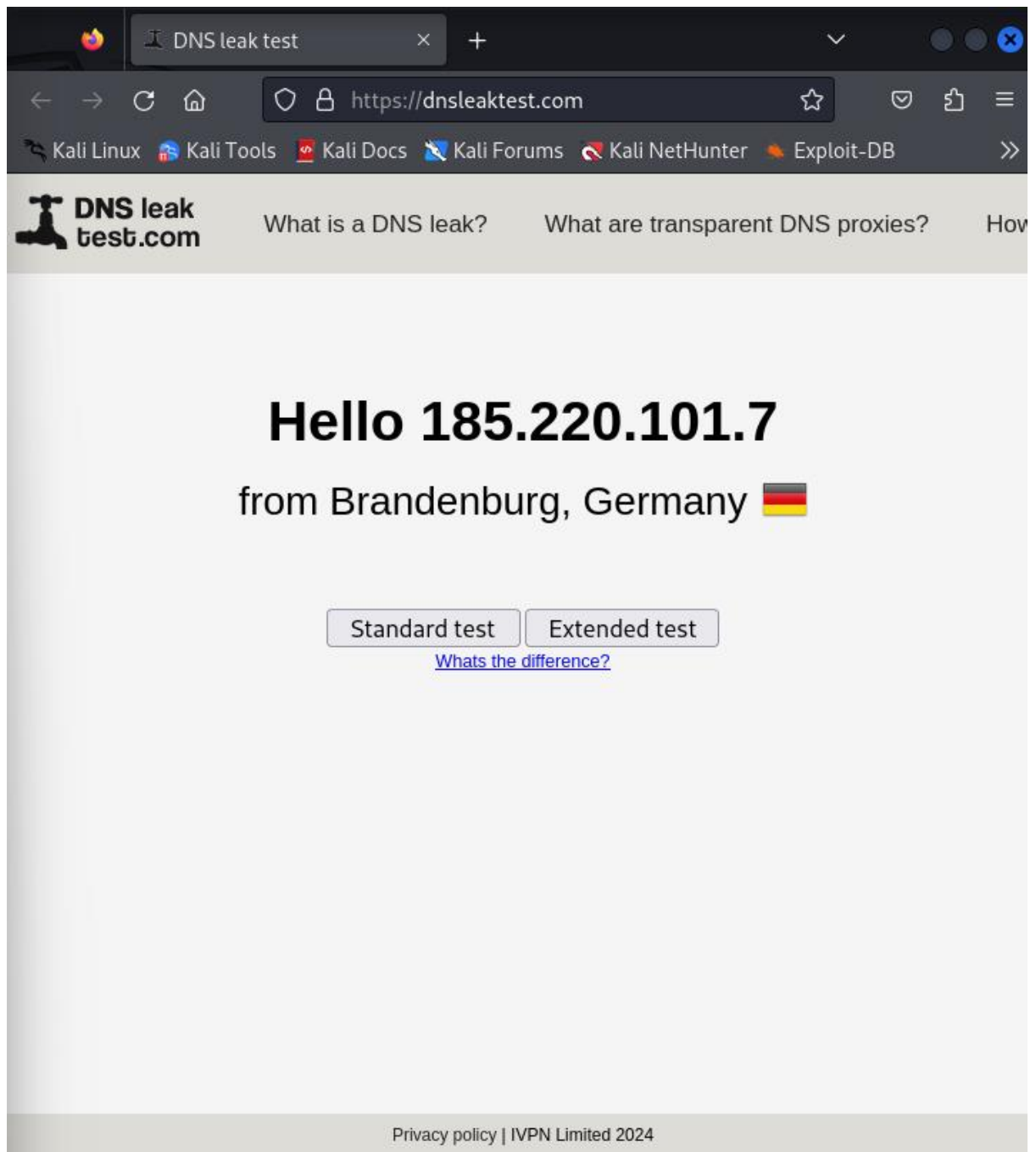
Tools Used: *(Mention the tools used in this project)*

- Kali Linux, Tor

Working Procedure: *(Answer in 200 words with bullet points)*

- To configure proxychains on kali we need to first install Tor in Kali linux using the following commands in terminal:
 - `sudo apt-get update`
 - `sudo apt-get install tor`
- Once tor is installed, start the service and enable it to start on boot
 - `sudo systemctl start tor`
 - `sudo systemctl enable tor`
- Verify if the tor is running using the following command
 - `systemctl status tor`
- Proxychains is already available in Kali Linux, if not install it using the following command:
 - `sudo apt-get install proxychains4`
- Edit the proxyChains configuration file to use Tor as the proxy. Access the file using the following command in the terminal:
 - `Sudo nano etc/proxychains4.conf`
- In the file , make the following changes:
 - Uncomment `dynamic_chain`
 - Comment `strict_chain`
 - Comment `random_chain`
 - Make sure that tor default port exists,i.e,
 - `Socks5 127.0.0.1 9050`

You can add more proxys to the file by searching them from "Socks5 Proxy List" online
- Save the file and Exit
- Finally to route firefox traffic through Tor, use the following command:
 - `proxychains firefox www.dnsleaktest.com`
- Firefox opens showing that your DNS requests are going through proxy
- The following is the outcome of the same:



• **Learning Outcomes: (Answer in 100 words with bullet points)**

- Learning to configure Proxychains to route network traffic through multiple proxies enhances anonymity
- Learned to setup and test various proxy servers, including SOCKS5, SOCKS4 and HTTP proxies
- Understood the way the proxy chains hides IP address and protect against tracking and surveillance :
 - Proxychains routes your network traffic through multiple proxy servers (SOCKS4, SOCKS5, and HTTP proxies) in a chain. This means your traffic goes through one proxy to another, creating multiple layers of anonymity.

- By chaining proxies, your IP address is hidden behind multiple proxy servers, making it much harder for websites or network sniffers to trace your original IP.
- The tool uses a configuration file (proxychains.conf) to define which proxies to use and in what order. It can be set up to use either a strict chain (each request goes through the same sequence of proxies) or a random chain (proxies are chosen randomly for each request).

Project - 5

Name of the Project: VPNs

Tools Used: *(Mention the tools used in this project)*

- Tor, VPN

Working Procedure: *(Answer in 200 words with bullet points)*

- Open Kali linux on your virtual machine .Open firefox or any other browser in kali and search for open vpn free and select enter.
- Select a VPN bundle from any of the trusted source and download it onto your system.
- Make sure to save the username and password for further use.
- After downloading the bundle , visit the path and extract the files from the path
- Now open the terminal and change directory in to the file path where you have extracted the files
- Run the following command for tcp :
 - sudo openvpn --config --your-config-file.ovpn
 - If you are not using the root terminal , you'll be prompted to enter the root(system) password along with vpn username and password
 - Make sure you type the right credentials and press enter
 - After initialization of VPN is complete, visit www.dnsleaktest.com to see that the host IP address and location being changed
- In www.dnsleaktest.com , below the system details , click on standard test button
- This will send queries to "your" server and provide the details of the Host IP, ISP, and location.
- For masking your ISP, type the command :
 - nano etc/resolv.conf
 - And change the name server to 1.1.1.1 (cloudflare)
 - And run dnsleaktest.com again in firefox
 - You'll notice that that IP address, location and ISP all of them have been completely proxied , protecting your identity
- For more anonymity , start tor in your terminal
- In the previous project where we used proxychains , use the same command and run dnsleaktest.com in firefox :
 - proxychains firefox www.dnsleaktest.com
- This will give you additional anonymity to your system

Learning Outcomes: *(Answer in 100 words with bullet points)*

- Learnt how to download ,install and configure VPN clients on Kali Linux

- Understood the process of connecting to a VPN server using OpenVPN with specific configuration files.
- Understood the importance of DNS leak tests to verify that your IP address and location are being masked.
- Learnt to analyze the results of DNS leak tests to ensure that the VPN is functioning correctly and that your identity is protected.
- Learnt to configure DNS settings by editing the `resolv.conf` file and changing the nameserver to enhance privacy and prevent DNS leaks.
- Developed the ability to use multiple tools (VPN, Tor, Proxychains) together to create an anonymity setup.